Cambridge
O Level

Cambridge International Examinations
Cambridge Ordinary Level

COMPUTER SCIENCE **2210/22**

Paper 2 **May/June 2016**

MARK SCHEME

Maximum Mark: 50

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge will not enter into discussions about these mark schemes.

Cambridge is publishing the mark schemes for the May/June 2016 series for most Cambridge IGCSE®, Cambridge International A and AS Level components and some Cambridge O Level components.

® IGCSE is the registered trademark of Cambridge International Examinations.

This syllabus is approved for use in England, Wales and Northern Ireland as a Cambridge International Level 1/Level 2 Certificate.

© UCLES 2016

CAMBRIDGE
International Examinations

Section A

1 (a) (i) Many correct answers, they must be meaningful. This is an example only.

– `NumSacks`, integer, number of sacks
– `SacksAccepted`, integer, number of sacks accepted
– `TotalWeight`, real, total weight of all sacks [3]

(ii) Any **three** from

– `TopWeight, 50.1`
– `BottomWeight 49.9`
– `TopWeightCement 25.1`
– `BottomWeightCement 24.9` [3]

(b) Any **five** from:

– initialise total weight of order
– input number of sacks for each type *outside loop(s)*
– loop for order completion
– input weight *inside loop(s)*
– add weight of accepted sack to total weight
– output total weight *outside all loop(s)*
– appropriate prompts for input number of sacks for each type and input weight [5]

Max 5 marks

Sample Answer 1

```
INPUT 'Number of sand sacks ordered ' num_sand_ordered
INPUT 'Number of cement sacks ordered ' num_cement_ordered
INPUT 'Number of gravel sacks ordered ' num_gravel_ordered
total_weight ← 0

FOR Counter ← 1 TO num_sand_ordered
   INPUT 'weight of sack of sand ' sack_weight
      total_weight ← total_weight + sack_weight
NEXT Counter
FOR Counter ← 1 TO num_cement_ordered
   INPUT 'weight of sack of cement ' sack_weight
      total_weight ← total_weight + sack_weight
NEXT Counter
FOR Counter ← 1 TO num_gravel_ordered
   INPUT 'weight of sack of gravel' sack_weight
      total_weight ← total_weight + sack_weight
NEXT Counter
Print 'Total weight of sacks is ' total_weight
```

© Cambridge International Examinations 2016

Sample Answer 2

```
INPUT 'Number of sand sacks ordered ' num_sand_ordered
INPUT 'Number of cement sacks ordered ' num_cement_ordered
INPUT 'Number of gravel sacks ordered ' num_gravel_ordered
total_sacks_ordered ← num_sand_ordered + num_cement_ordered +
num_gravel_ordered
total_weight ← 0

FOR Counter ← 1 TO total_sacks_ordered
   INPUT 'weight of sack ' sack_weight
      total_weight ← total_weight + sack_weight
NEXT Counter
Print 'Total weight of sacks is ' total_weight
```

(c) (i) **1** mark for value reason, all values and reasons must be different. There are many possible correct answers these are examples only.

Data value 49.95
Reason – normal data sand should be accepted

Data value 50.1
Reason – boundary data sand should be rejected [2]

(ii)

Data value 25
Reason – normal data cement should be accepted

Data value 26.7
Reason – abnormal data cement that should be rejected [2]

(d) Maximum **5** marks in total, maximum **3** marks if only programming statements used

Description (may include reference to program statements)

– use of prices for calculation of regular price either numbers, variables or constants (sand and gravel 2, cement 3)

– description of calculation of regular price, multiply no of sand sacks by 2, multiply no of gravel sacks by 2, multiply no of cement sacks by 3

– output of regular price

– description of calculating the number of special packs using the pack information (2 sacks of sand, 2 sacks of gravel and 1 sack of cement)

– …….repeat until there are no more packs in the order (less than 2 sacks of sand or less than 2 sacks of gravel or no sacks of cement

– calculation of discount price and/or amount saved

– output discount price and/or amount saved [5]

© Cambridge International Examinations 2016

Section B

2 (i) **1** mark for each improvement

use `FOR … NEXT` instead of `REPEAT … UNTIL`
Move `PRINT` to after the end of the loop
Add error checking to check that the value input is positive [3]

(ii) **3** marks maximum, **1** mark for each improvement correctly included.

Sample answer below

```
1 Total = 0
2   FOR Counter = 1 To 10
3    REPEAT
4     INPUT Num
5   UNTIL Num >0
6     Total = Total + Num
7 NEXT Counter
8 PRINT Total
```

[3]

3

Area	Tins	Height	Width	Doors	Windows
0	0	3	5	1	0
13.5		3	7	0	0
34.5		3	5	0	3
46.5		3	7	1	1
65		–1	0	0	0
	7				

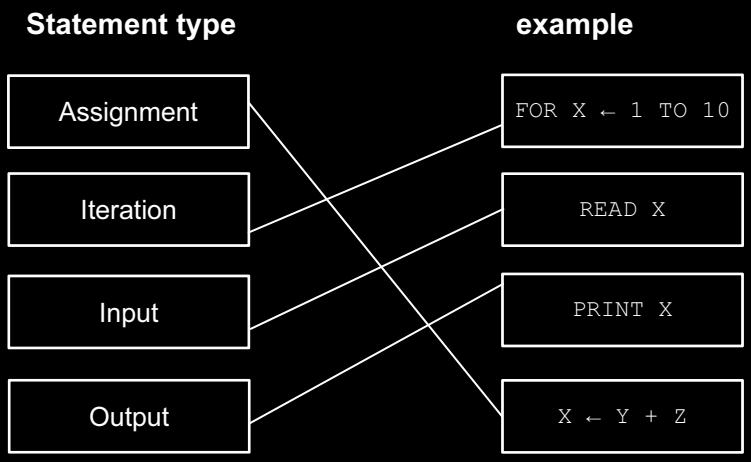
(2 marks) ←(1 mark)→ ← (1 mark) →

1 mark 0, 13.5
1 mark for rest

[4]

© Cambridge International Examinations 2016

4 **1** mark for each correct line, maximum **3** (zero correct **0**, one correct **1**, two correct **2**, three or four correct **3**), each box must have only one connection.



[3]

5 – data structure (one—dimensional) array …….
– …… reason to simplify programming/ make programs shorter, etc. [2]

6 – IF (… THEN … ELSE … ENDIF)

– CASE (… OF … OTHERWISE … ENDCASE) [2]

7 **(a)** – 7 [1]

(b) – Brochure Number……..
– ….. Uniquely identifies each record/each Brochure Number different/no duplicates [2]

(c) – Number of Seats – number/integer
– Price in $ – currency/real [2]

(d) **1** mark for each correct result, **1** mark for the results in descending order of price

– Recliner sofa 1,200 RS23
– Recliner chair 600 RC01 [3]

© Cambridge International Examinations 2016

(e)

Field:	Brochure Number	Material	Colour	Price in $	Number of Seats
Table:	SOFASELECT	SOFASELECT	SOFASELECT	SOFASELECT	SOFASELECT
Sort:					
Show:	☑	☑	☑	☑	☐
Criteria:					>2
or:					
	(1 mark)	(1 mark)	(1 mark)	(1 mark)	(1 mark)

[5]

© Cambridge International Examinations 2016